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Student Number



Barker
College

2022

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

Section I - Multiple Choice

Write your student number at the top of this page.

Select the alternative A, B C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ A 2 B 6 C 8 D 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

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Start
Here →

- | | |
|--|---|
| 1. A ☐ B ☐ C ☐ D ☐ | 6. A ☐ B ☐ C ☐ D ☐ |
| 2. A ☐ B ☐ C ☐ D ☐ | 7. A ☐ B ☐ C ☐ D ☐ |
| 3. A ☐ B ☐ C ☐ D ☐ | 8. A ☐ B ☐ C ☐ D ☐ |
| 4. A ☐ B ☐ C ☐ D ☐ | 9. A ☐ B ☐ C ☐ D ☐ |
| 5. A ☐ B ☐ C ☐ D ☐ | 10. A ☐ B ☐ C ☐ D ☐ |

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Student Number

2022

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

Staff Involved:

BHC*
GDH
KJL

35 copies

8:30 AM
FRIDAY 5 AUGUST

TOPICS COVERED

- Complex Numbers I & II
- Proof
- Integration
- Vectors
- Mechanics

General

Instructions:

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

Total marks:
100

Section I – 10 marks (pages 4-8)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 9-14)

- Attempt Questions 11-16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Given that x and y are **positive integers**, which one of the following statements is false?

A. $\forall x \exists y \left(\frac{2x}{y} = 1 \right)$

B. $\forall y \exists x \left(\frac{x}{y} = 2 \right)$

C. $\forall x \exists y \left(\frac{x}{2y} = 1 \right)$

D. $\forall y \exists x \left(\frac{y}{x} = \frac{1}{2} \right)$

2 Let $z = \frac{\sqrt{3} - i}{1 - i}$.

What are the modulus and argument, respectively, of z ?

A. $\sqrt{2}$ and $\frac{\pi}{12}$

B. $\sqrt{2}$ and $-\frac{\pi}{12}$

C. $\frac{1}{\sqrt{2}}$ and $\frac{\pi}{12}$

D. $\frac{1}{\sqrt{2}}$ and $-\frac{\pi}{12}$

- 3 What is the Cartesian equation of the line with vector equation

$$r = \begin{pmatrix} 8 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix} ?$$

- A. $2x + 3y = 36$
 B. $3x + 2y = 36$
 C. $2x - 3y = 36$
 D. $3y - 2x = 36$

- 4 In the complex plane, $2 + i$ and $6 - 9i$ represent the endpoints of the diameter of a circle.
 What is the equation of the circle?

- A. $|z - 4 - 4i| = \sqrt{29}$
 B. $|z - 4 - 4i| = 29$
 C. $|z - 4 + 4i| = \sqrt{29}$
 D. $|z - 4 + 4i| = 29$

- 5 A particle is moving in simple harmonic motion on a straight line.

At time t seconds, it has displacement x metres from a fixed point on the line and velocity v m/s given by $v^2 = 4 - \frac{1}{3}x^2 + \frac{4}{3}x$.

What is the period of the motion?

- A. 2π seconds
 B. $\frac{2\pi}{3}$ seconds
 C. $\frac{2\pi}{\sqrt{3}}$ seconds
 D. $2\sqrt{3}\pi$ seconds

- 6 Which one of the following is an expression for $\int \frac{\cos^3 x + \sin^3 x}{\cos x + \sin x} dx$?

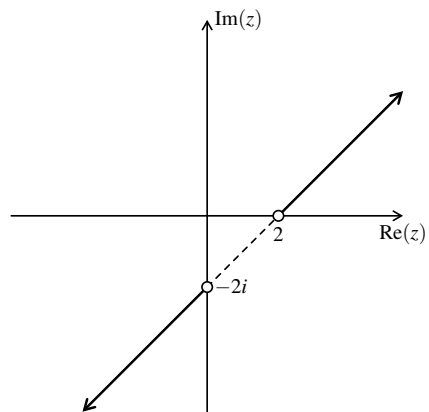
- A. $x + \frac{1}{4} \cos 2x + C$
 B. $x - \frac{1}{4} \cos 2x + C$
 C. $x + \frac{1}{2} \cos 2x + C$
 D. $x - \frac{1}{2} \cos 2x + C$

- 7 A particle of mass m falls vertically from rest under gravity in a medium in which the resistance to motion has a magnitude $\frac{1}{30}mv^2$ where v m/s is the speed of the particle and $g = 9.8 \text{ m/s}^2$ is the acceleration due to gravity.

What is the terminal velocity of the particle?

- A. 294 m/s
 B. 9.8 m/s
 C. 30 m/s
 D. 17.1 m/s

- 8 The path traced out by a complex number z is shown by the solid rays in the Argand diagram below.



Which one of the following is the equation of the path traced by z ?

- A. $\text{Arg}(z - 2) - \text{Arg}(z + 2i) = \pi$
- B. $\text{Arg}(z + 2i) - \text{Arg}(z - 2) = \frac{\pi}{4}$
- C. $\text{Arg}(z - 2) - \text{Arg}(z + 2i) = 0$
- D. $\text{Arg}(z + 2i) - \text{Arg}(z - 2) = \pi$

- 9 Without evaluating the integral, which one of the following must be negative?

- A. $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{1 + \cos x}{x} dx$
- B. $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\tan x}{x^3} dx$
- C. $\int_{-1}^1 (x^2 - 1) e^{-x^2} dx$
- D. $\int_{-1}^1 \sin(x^3) dx$

- 10 Let P be a polynomial with real coefficients. If $P(\bar{z}) = 3 + i$, what is the value of $P(z)$?

- A. $3 + i$
- B. $3 - i$
- C. $-1 + 3i$
- D. $-1 - 3i$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks) Use a SEPARATE Writing Booklet

- (a) The quadratic equation $z^2 + pz + q = 0$, where p and q are real, has a root of $\sqrt{2} + i$. Find the values of p and q . 2

- (b) Evaluate $\int_0^1 \tan^{-1} x \, dx$. 2

- (c) Find, correct to the nearest degree, the angle between the vectors $\underline{u}$ and $\underline{v}$, given that $\underline{u} + \underline{v} = 3\underline{i} - 3\underline{j} + 5\underline{k}$ and $\underline{u} - \underline{v} = \underline{i} + \underline{j} + \underline{k}$. 3

- (d) A subset of the complex plane is described by the relation $|z - (2 + 4i)| = 1$.
(i) Draw a neat sketch of this relation. 1

- (ii) Given that z is a complex number that satisfies the relation, find the minimum and maximum values of $|z|$. Give your answers in exact form. 2

- (iii) Given that z is a complex number that satisfies the relation, find the minimum and maximum values of $\text{Arg}(z)$, correct to the nearest degree. 2

- (e) The equations of lines ℓ_1 and ℓ_2 are given with respect to a fixed origin. 2

$$\ell_1: \underline{r}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \quad \text{and} \quad \ell_2: \underline{r}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} p \\ 4 \\ 3 \end{pmatrix}$$

where λ and μ are parameters and p is a constant.

What is the value of p if ℓ_1 is perpendicular to ℓ_2 ?

Question 12 (16 marks) Use a SEPARATE Writing Booklet

- (a) At time t , a particle has the position:
 $x = 3t^2$, $y = t^2 - 2t$, $z = t^3$

- (i) Find the magnitude of the velocity of the particle at time $t = 1$. 2

- (ii) Find the vector that is the component of the velocity at time $t = 1$ in the direction of $\underline{u} = 2\underline{i} - \underline{j} + 2\underline{k}$. 2

- (b) Consider the conditional statement S , where n is a positive integer:

S : “If n is not divisible by 12, then n is not divisible by 3.”

- (i) Write, and clearly label, the converse, the contrapositive and the negation of S . 3

- (ii) Determine whether the converse and the contrapositive of S are true or false, briefly justifying each answer. 2

- (c) Let the point A in the complex plane represent $z = 4 + 6i$, where O is the origin.

Let C be the new position of $\vec{OA}$ after it has been rotated 120° clockwise and its magnitude increased by a factor of 4.

- (i) Find the point C in the form $(a + b\sqrt{c}) + i(d + e\sqrt{f})$. 2

- (ii) Find the point B in the form $x + iy$, such that $OABC$ is a parallelogram. 1

- (d) A particle is undergoing simple harmonic motion about the origin with period $\frac{\pi}{4}$ seconds. Initially, the particle is at rest 5 metres to the right of the origin. 4

Find the first two times when the particle's speed is half its maximum speed.

Question 13 (15 marks) Use a SEPARATE Writing Booklet

- (a) Note that each of parts (α), (β) and (γ) use the following definition of z^2 , but they can be solved independently of each other.

Let $z^2 = 5 + 5i$.

(α) Find z and express it in the form $x + iy$, where x and y are real numbers. **3**

(β) If $z^2 w = z^2 + \overline{z^2} \sqrt{2}$, find w and express it in the form $x + iy$. **3**

(γ) If iz is a solution to $x^m + q = 0$, where m and q are positive integers, find a possible value of m and q . **3**

- (b) A particle moves in a straight line. At time t seconds, its displacement from a fixed origin is x metres and its velocity is v m/s.

The acceleration of the particle is given by $\ddot{x} = 2x - 4\sqrt{2}$.

At time $t = 0$, the particle is at the origin with $v = 4$ m/s.

(i) Show that $v = 4 - x\sqrt{2}$. **3**

(ii) Find an expression for x in terms of t . **3**

Question 14 (16 marks) Use a SEPARATE Writing Booklet

- (a) A particle is projected from a fixed origin, O .

There are two forces acting on the particle: gravity and air resistance.

The particle's displacement is measured in metres, and time is measured in seconds.

(i) The initial horizontal component of the velocity is u . Its subsequent horizontal motion is modelled by $\frac{d\dot{x}}{dt} = -k\dot{x}$. **3**

Show that the particle's horizontal displacement is given by $x = \frac{u}{k} (1 - e^{-kt})$.

(ii) The initial vertical component of the velocity is w . Its subsequent vertical motion is modelled by $\frac{d\dot{y}}{dt} = -k\dot{y} - g$ where g is the acceleration due to gravity. **3**

Show that the particle's vertical displacement is given by

$$y = \frac{kw + g}{k^2} (1 - e^{-kt}) - \frac{gt}{k}.$$

(iii) Show that the Cartesian equation of motion is given by **2**

$$y = \left(\frac{kw + g}{ku} \right) x + \frac{g}{k^2} \ln \left(1 - \frac{kx}{u} \right).$$

(iv) Suppose that $u = 9$ m/s, $w = 6$ m/s, $k = 0.2$ and $g = 10$ m/s². Determine whether the particle will pass over a wall of height 0.15m at a horizontal distance of 9 m from the origin. **1**

(b) (i) Prove that $\forall k \in \mathbb{Z}^+$, if k^4 is even, then k is even. **2**

(ii) Prove that $\sqrt[4]{8}$ is irrational. **3**

(c) If $a, b \in \mathbb{R}^+$ and $a + b = 8$, show that $\frac{1}{a} + \frac{1}{b} \geq \frac{1}{2}$. **2**

Question 15 (15 marks) Use a SEPARATE Writing Booklet

(a) Use an appropriate substitution to find $\int \frac{dx}{x\sqrt{1-x}}$. 3

(b) Let $z = e^{i\theta}$. 1

(i) Show that $z^n + \frac{1}{z^n} = 2\cos(n\theta)$. 1

(ii) Show that $\left(z + \frac{1}{z}\right)^5 = \left(z^5 + \frac{1}{z^5}\right) + 5\left(z^3 + \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right)$. 1

(iii) Hence, find $\int \cos^5 \theta \, d\theta$. 2

(c) Prove by mathematical induction that 4

$$\frac{d^n}{dx^n}(\sqrt{x}) = \left[\frac{(-1)^{n-1}(2n-2)!}{2^{2n-1}(n-1)!} \right] x^{\frac{-(2n-1)}{2}} \text{ for integers } n \geq 2,$$

where $\frac{d^n}{dx^n}(\sqrt{x})$ is the n th derivative of $\sqrt{x}$.

(d) Let m and n be positive integers. 4

Use proof by contradiction to show that there is no positive prime number p such that $25m^2 + 30m + 9 = n^2 + p$.

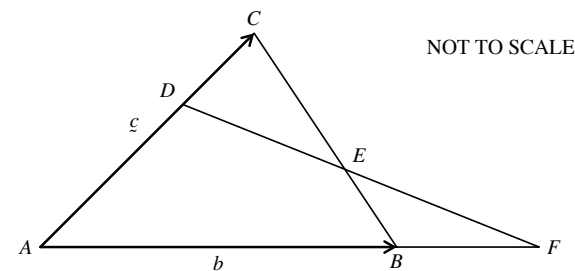
Question 16 (14 marks) Use a SEPARATE Writing Booklet

(a) In $\triangle ABC$ below, D is the point on AC such that $AD:DC = 3:1$.

E is the point on BC such that $BE:EC = 1:3$.

When DE is extended, it meets the extension of AB at F .

Let $\vec{AB} = \underline{b}$ and $\vec{AC} = \underline{c}$.



(i) Show that $\vec{DE} = \frac{3}{4}\underline{b} - \frac{1}{2}\underline{c}$. 2

(ii) Find the ratio of $AF:BF$. 3

(b) The point $A(-2, 4, 9)$ is reflected to the point B in the line defined by the equation 3

$$\underline{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \text{ where } \lambda \in \mathbb{R}.$$

Find the coordinates of B .

(c) (i) Find $\int \sin^4 x \cos^3 x \, dx$. 2

(ii) Use integration by parts to prove that 3

$$\int \sin^m x \cos^n x \, dx = \frac{\sin^{m+1} x \cos^{n-1} x}{m+n} + \frac{n-1}{m+n} \int \sin^m x \cos^{n-2} x \, dx.$$

(iii) Use the reduction formula in part (ii) to confirm your answer for part (i). 1

End of paper

Barker Mathematics Extension 2 Solutions 5.8.22

Multiple Choice Questions

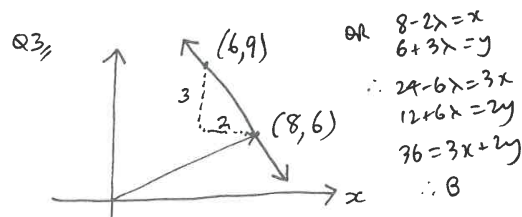
Q1, C is false when $x=1$

$$Q2, z = \frac{\sqrt{3}-i}{1-i} = \frac{2 \operatorname{cis}(-\frac{\pi}{6})}{\sqrt{2} \operatorname{cis}(-\frac{\pi}{4})}$$

$$\text{So } |z| = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\text{and } \arg(z) = -\frac{\pi}{6} - (-\frac{\pi}{4}) = \frac{\pi}{12}$$

Hence A



The equation of the line passing through (8,6) and (6,9) is

$$y - 9 = -\frac{3}{2}(x - 6)$$

$$2y - 18 = -3x + 18$$

$$3x + 2y = 36$$

Hence B

Q4, The midpoint of the interval joining $(2+i)$ to $(6-9i)$ is $(4-4i)$, the centre of the circle. The distance from $(2+i)$ to $(4-4i)$ is $\sqrt{29}$, the radius of the circle. The equation is $|z - (4-4i)| = \sqrt{29}$

Hence C.

$$Q5, \text{ Period} = \frac{2\pi}{n}$$

$$\frac{d}{dx}(\frac{1}{2}v^2) = \frac{d}{dx}(4 - \frac{1}{6}x^2 + \frac{2}{3}x)$$

$$= -\frac{2}{6}x + \frac{2}{3}$$

$$= -\frac{1}{3}(x-2) \therefore n^2 = \frac{1}{3}$$

$$\therefore n = \frac{1}{\sqrt{3}}$$

$$\therefore \text{Period} = \frac{2\pi}{(\frac{1}{\sqrt{3}})} = 2\sqrt{3} \pi \text{ seconds}$$

Hence D.

$$Q6, \cos^3 x + \sin^3 x$$

$$= (\cos x + \sin x) \times (\cos^2 x - \cos x \sin x + \sin^2 x)$$

$$\therefore I = \int \cos^2 x - \cos x \sin x + \sin^2 x dx$$

$$I = \int 1 - \cos x \sin x dx$$

$$I = x + \frac{1}{2}(\cos x)^2 + C$$

$$I = x + \frac{1}{2}\left(\frac{1+\cos 2x}{2}\right) + C$$

$$I = x + \frac{1}{4}\cos 2x + C$$

Hence A.

$$\text{or } I = \int 1 - \frac{\sin 2x}{2} dx$$

$$= x + \frac{\cos 2x}{4} + C$$

$$Q7, \downarrow mg \quad \uparrow (-) \frac{1}{30}mv^2$$

$$m\ddot{z} = mg - \frac{1}{30}mv^2$$

$$\ddot{z} = g - \frac{1}{30}v^2$$

when terminal velocity is achieved $\ddot{z} = 0$ so

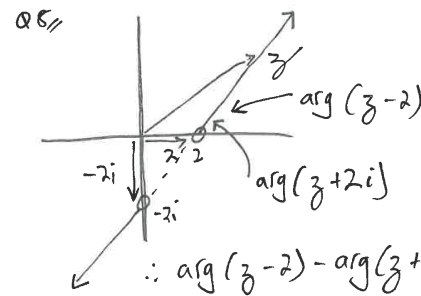
$$\text{solve } 0 = 9.8 - \frac{1}{30}v^2$$

$$v^2 = 9.8 \times 30$$

$$v^2 = 294$$

$$v = 17.1 \text{ m/sec}$$

Hence D.



$$\therefore \arg(z-2) - \arg(z+2i) = 0 - 0 = 0$$

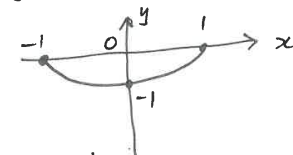
Hence C.

Q9, which integral must be a negative definite integral?

The integral in A is an odd function and $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \text{odd function} = 0$

The integral in C is an even function and the graph of $y = (x^2-1)e^{-x^2}$ lies below the x axis between 1 and -1. In fact:

$$y = (x^2-1)e^{-x^2}$$



Hence $\int_{-1}^1 \text{even function} = \text{negative answer}$

Hence C.

D is $\int_{-a}^a \text{odd function} = 0$

B is $\int_{-a}^a \text{even function} dx$ whose graph is above the x axis.

In addition, both B & A have asymptotes at $x=0$, so we can't integrate across the asymptote anyway.

Q10 Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 x^0$ where $a_0, \dots, a_n$ real coefficients

$$\therefore p(\bar{z}) = a_n (\bar{z})^n + a_{n-1} (\bar{z})^{n-1} + \dots + a_1 (\bar{z}) + a_0 (\bar{z})^0$$

$$\text{Now } (\bar{z})^n = \overline{(z^n)}$$

$$\therefore p(\bar{z}) = a_n \overline{(z^n)} + a_{n-1} \overline{(z^{n-1})} + \dots + a_1 \overline{(z)} + a_0 \overline{(z^0)}$$

$$= \overline{a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 z^0} \quad \text{since } a_0, \dots, a_n \text{ real}$$

$$= \overline{a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 z^0} \quad \text{since } \overline{a+b} = \overline{a} + \overline{b}$$

$$= \overline{p(z)}$$

$$\therefore 3+i = \overline{p(z)}$$

$$\therefore 3-i = p(z) \quad \therefore B$$

Q11

(a) Since the coefficients of all terms are real, then roots occur in conjugate pairs. Hence $\alpha = \sqrt{2} + i$ and $\beta = \sqrt{2} - i$, so

$$\alpha + \beta = 2\sqrt{2} = -\frac{b}{a} = -p$$

$$\text{Hence } p = -2\sqrt{2}$$

$$\alpha\beta = (\sqrt{2} + i)(\sqrt{2} - i)$$

$$= 2 + 1$$

$$= \frac{c}{a}$$

$$= q \quad \text{Hence } q = 3$$

$$(b) \int_0^1 \tan^{-1} x \, dx$$

$$= x \tan^{-1} x \Big|_0^1 - \int_0^1 x \cdot \frac{1}{1+x^2} \, dx$$

$$= \frac{\pi}{4} - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} \, dx$$

$$= \frac{\pi}{4} - \frac{1}{2} [\ln(1+x^2)]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$(c) \underline{u} + \underline{v} = 3\underline{i} - 3\underline{j} + 5\underline{k}$$

$$\underline{u} - \underline{v} = \underline{i} + \underline{j} + \underline{k}$$

$$\therefore 2\underline{u} = 4\underline{i} - 2\underline{j} + 6\underline{k}$$

$$\underline{u} = 2\underline{i} - \underline{j} + 3\underline{k}$$

$$\text{Hence } \underline{v} = \underline{i} - 2\underline{j} + 2\underline{k}$$

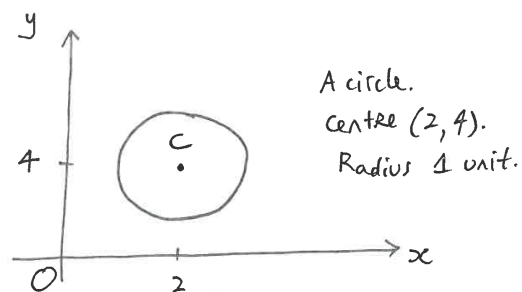
$$\text{Then } \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$= \frac{2+2+6}{\sqrt{14} \times 3}$$

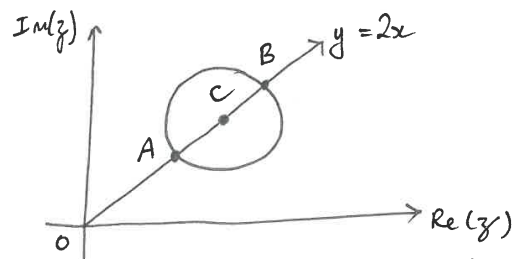
$$= \frac{10}{3\sqrt{14}}$$

$$\therefore \theta = 27^\circ \quad (\text{nearest degree})$$

(d)(i) $|z - (2+4i)| = 1$

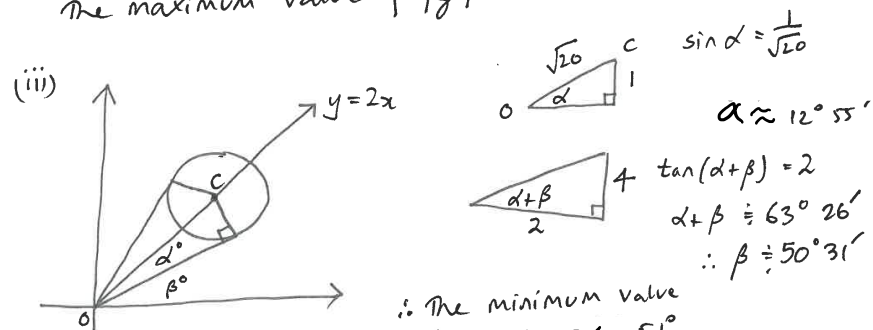


(ii) The centre of the circle is C.
The length of OC is $\sqrt{4^2 + 2^2} = \sqrt{20}$.



The minimum value of $|z|$ is $OA = \sqrt{20} - 1 = 2\sqrt{5} - 1$

The maximum value of $|z|$ is $OB = \sqrt{20} + 1 = 2\sqrt{5} + 1$



$\therefore$ The minimum value of $\text{Arg}(z) \approx 51^\circ$

The maximum value of

$$\text{Arg}(z) = \beta + 2\alpha = 50^\circ 31' + 2 \times 12^\circ 55' \approx 76^\circ 21' \approx 76^\circ$$

(e) Now l_1 is perpendicular to l_2 if

$$\begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} p \\ q \\ 3 \end{pmatrix} = 0$$

$$\therefore -2p + 4 + 12 = 0$$

$$-2p = -16$$

$$p = 8$$

Q12, (a) (i)

$$\left. \begin{aligned} \frac{dx}{dt} &= 6t \\ \frac{dy}{dt} &= 2t - 2 \\ \frac{dz}{dt} &= 3t^2 \end{aligned} \right\} \text{When } t=1, \begin{aligned} \frac{dx}{dt} &= 6 \\ \frac{dy}{dt} &= 0 \\ \frac{dz}{dt} &= 3 \end{aligned}$$

$$\therefore |\underline{v}| = \sqrt{6^2 + 0^2 + 3^2} = \sqrt{45} = 3\sqrt{5} \text{ m/s}$$

(ii) Now $\underline{v} = \begin{pmatrix} 6 \\ 0 \\ 3 \end{pmatrix}$ and $\underline{u} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ then

$$\begin{aligned} \text{proj}_{\underline{u}} \underline{v} &= \frac{12 + 0 + 6}{(\sqrt{4 + 1 + 4})^2} \underline{u} \\ &= 2 \underline{u} \\ &= 2 \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix} \end{aligned}$$

Q12, (b) S: "If n is not divisible by 12, then n is not divisible by 3."

(i) The converse is:

"If n is not divisible by 3, then n is not divisible by 12."

The contrapositive is:

"If n is divisible by 3, then n is divisible by 12."

the negation is:

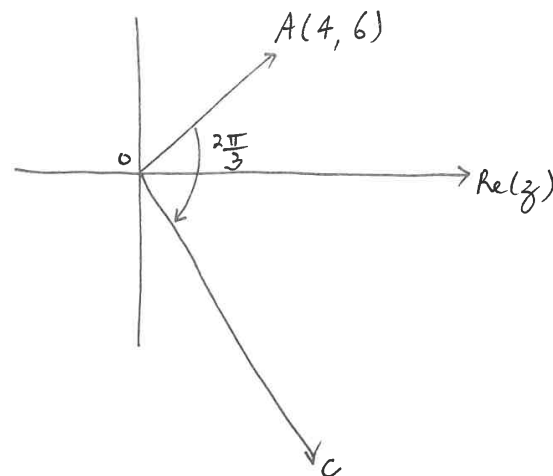
"There exists an integer n with the property that n is not divisible by 12, but n is divisible by 3."

(ii) The contrapositive is false. That is because the contrapositive has the same truth value as S , and S is false. [S is false because 6 is not divisible by 12, but 6 is divisible by 3.]

Alternatively; the contrapositive is false by this counterexample: 6 is divisible by 3, but 6 is not divisible by 12.

The converse is true. That is because if the prime factorisation of n does not have a 3 in it, then n will not be divisible by 12.

Q12, (c)



$$\begin{aligned}
 \text{i) } \vec{OC} &= z \times 4 e^{-2\pi/3 i} \\
 &= (4+6i) \times 4 \left(\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \right) \\
 &= (4+6i) \times 4 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \\
 &= (4+6i) (-2-2\sqrt{3} i) \\
 &= -8 + 12\sqrt{3} - 12i - 8\sqrt{3} i \\
 &= (12\sqrt{3} - 8) + i(-12 - 8\sqrt{3}) = (-8 + 12\sqrt{3}) + i(-12 - 8\sqrt{3})
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \vec{OB} &= \vec{OA} + \vec{OC} \\
 &= (4+6i) + (12\sqrt{3} - 8) + i(-12 - 8\sqrt{3}) \\
 &= (12\sqrt{3} - 4) + i(-6 - 8\sqrt{3})
 \end{aligned}$$

(d) Period = $\frac{\pi}{4}$ seconds

So $\frac{2\pi}{n} = \frac{\pi}{4} \therefore n = 8$

The equation of motion is

$$x = 5 \cos 8t$$

Hence $v = -40 \sin 8t$

So the maximum speed is 40 m/sec

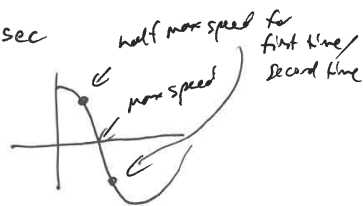
Need to find t when $v = -20$

$$-20 = -40 \sin 8t$$

$$\sin 8t = \frac{1}{2}$$

$$8t = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore t = \frac{\pi}{48}, \frac{5\pi}{48} \text{ seconds}$$



Q13 (a) let $z^2 = 5+5i$

$$\begin{aligned} (\alpha) \quad (x+iy)^2 &= 5+5i \\ x^2-y^2 &= 5 \quad (1) \quad 2xy = 5i \\ 2xy &= 5 \quad (2) \end{aligned}$$

$$\begin{aligned} \text{Now } (x^2+y^2)^2 &= (x^2-y^2)^2 + 4x^2y^2 \\ &= 5^2 + 5^2 \\ &= 50 \end{aligned}$$

$$\therefore x^2+y^2 = \pm 5\sqrt{2}$$

But $x^2 \geq 0$ and $y^2 \geq 0$ so

$$x^2+y^2 = 5\sqrt{2} \quad (3)$$

Now (1) + (3) gives

$$\begin{aligned} 2x^2 &= 5+5\sqrt{2} \\ x &= \pm \sqrt{\frac{5+5\sqrt{2}}{2}} \end{aligned}$$

(3) - (1) gives

$$\begin{aligned} 2y^2 &= 5\sqrt{2}-5 \\ y &= \pm \sqrt{\frac{5\sqrt{2}-5}{2}} \end{aligned}$$

From $2xy = 5$ we know x and y are the same sign. Hence

$$x+iy = \pm \left(\sqrt{\frac{5+5\sqrt{2}}{2}} + i \sqrt{\frac{5\sqrt{2}-5}{2}} \right)$$

(β) Given that $z^2 w = z^2 + \bar{z}^2 \sqrt{2}$, to find w .

$$(5+5i)w = (5+5i) + (5-5i)\sqrt{2}$$

$$(1+i)w = (1+i) + (1-i)\sqrt{2}$$

$$(1+i)w = (1+\sqrt{2}) + (1-\sqrt{2})i$$

$$w = \frac{(1+\sqrt{2}) + (1-\sqrt{2})i}{(1+i)} \times \frac{(1-i)}{(1-i)}$$

$$w = \frac{(1+\sqrt{2}) + (1-\sqrt{2})i - (1+\sqrt{2})i + (1-\sqrt{2})}{2}$$

$$w = \frac{2 - 2\sqrt{2}i}{2}$$

$$w = 1 - \sqrt{2}i$$

(γ) $z^2 = 5+5i = 5\sqrt{2} e^{i\frac{\pi}{4}}$

$$\therefore z = \pm \sqrt{5\sqrt{2}} e^{i(\frac{\pi}{8})}$$

$$\therefore iz = \pm \sqrt{5\sqrt{2}} e^{i(\frac{\pi}{8} + \frac{\pi}{2})}$$

$$iz = \pm \sqrt{5\sqrt{2}} e^{i(\frac{5\pi}{8})}$$

$$\begin{aligned} (iz)^8 &= (5\sqrt{2})^4 e^{i5\pi} \\ &= 5^4 \times 2^2 \times (\cos 5\pi + i \sin 5\pi) \\ &= -2500 \end{aligned}$$

$\therefore$ Possible values of m and n are
 $m = 8$ and $n = 2500$

Q13 (b)

$$(i) \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x - 4\sqrt{2}$$

$$\frac{1}{2} v^2 = x^2 - 4\sqrt{2}x + C$$

But when $x=0$, $v=4$ so $C=8$.

$$v^2 = 2x^2 - 8\sqrt{2}x + 16$$

$$v^2 = 2(x^2 - 4\sqrt{2}x + 8)$$

$$v^2 = 2(x - 2\sqrt{2})^2$$

$$v = \pm \sqrt{2} (x - 2\sqrt{2})$$

But when $x=0$, $v=4$ so

$$v = -\sqrt{2} (x - 2\sqrt{2})$$

$$\text{Hence } v = 4 - x\sqrt{2},$$

as required.

$$(ii) \text{ Now } v = \frac{dx}{dt} = 4 - x\sqrt{2}$$

$$\int \frac{dx}{4 - \sqrt{2}x} = \int dt$$

$$-\frac{1}{\sqrt{2}} \ln |4 - \sqrt{2}x| + C = t$$

But when $t=0$, $x=0$ so

$$C = \frac{1}{\sqrt{2}} \ln(4)$$

$$\therefore t = \frac{1}{\sqrt{2}} \ln(4) - \frac{1}{\sqrt{2}} \ln |4 - \sqrt{2}x|$$

$$t = \frac{1}{\sqrt{2}} \ln \left| \frac{4}{4 - \sqrt{2}x} \right|$$

$$e^{\sqrt{2}t} = \frac{4}{4 - \sqrt{2}x}$$

$$\rightarrow 4 - \sqrt{2}x = 4e^{-\sqrt{2}t}$$

$$4 - 4e^{-\sqrt{2}t} = \sqrt{2}x$$

$$x = \frac{4 - 4e^{-\sqrt{2}t}}{\sqrt{2}}$$

$$x = 2\sqrt{2} - 2\sqrt{2}e^{-\sqrt{2}t}$$

$$x = 2\sqrt{2}(1 - e^{-\sqrt{2}t})$$

Q14 (a)

$$(i) \frac{dx}{dt} = -kx$$

$$\int \frac{1}{x} dx = \int -k dt$$

$$\ln(x) = -kt + C$$

But $\dot{x} = u$ when $t=0$

$$\text{so } C = \ln u$$

$$\therefore \ln x = -kt + \ln u$$

$$\ln \left(\frac{x}{u} \right) = -kt$$

$$\frac{x}{u} = e^{-kt}$$

$$x = ue^{-kt}$$

$$\therefore x = \int ue^{-kt} dt$$

$$x = -\frac{u}{k} e^{-kt} + H$$

When $t=0$, $x=0$ so

$$H = \frac{u}{k}$$

$$\therefore x = \frac{u}{k} - \frac{u}{k} e^{-kt}$$

$$x = \frac{u}{k} (1 - e^{-kt})$$

as required.

$$(ii) \frac{dy}{dt} = -ky - g$$

$$= -k \left(y + \frac{g}{k} \right)$$

$$\int \frac{dy}{y + \frac{g}{k}} = \int -k dt$$

$$\ln \left(y + \frac{g}{k} \right) = -kt + C$$

When $t=0$, $y=w$ so

$$C = \ln \left(w + \frac{g}{k} \right)$$

$$\therefore -kt = \ln \left(\frac{y + \frac{g}{k}}{w + \frac{g}{k}} \right)$$

$$\left(w + \frac{g}{k} \right) e^{-kt} = y + \frac{g}{k}$$

$$y = \left(w + \frac{g}{k} \right) e^{-kt} - \frac{g}{k}$$

$$\therefore y = \int \left(\left(w + \frac{g}{k} \right) e^{-kt} - \frac{g}{k} \right) dt$$

$$y = -\frac{1}{k} \left(w + \frac{g}{k} \right) e^{-kt} - \frac{g}{k} t + F$$

But when $t=0$, $y=0$ so

$$F = \frac{1}{k} \left(w + \frac{g}{k} \right)$$

$$\therefore y = -\frac{1}{k} \left(w + \frac{g}{k} \right) e^{-kt} - \frac{g}{k} t + \frac{1}{k} \left(w + \frac{g}{k} \right)$$

$$y = \frac{w}{k} (1 - e^{-kt}) + \frac{g}{k^2} (1 - e^{-kt}) - \frac{g}{k} t$$

$$y = \frac{wk}{k^2} (1 - e^{-kt}) + \frac{g}{k^2} (1 - e^{-kt}) - \frac{g}{k} t$$

$$y = \frac{kw + g}{k^2} (1 - e^{-kt}) - \frac{g}{k} t$$

as required.

(iii) From part (i),

$$x = \frac{\mu}{k} (1 - e^{-kt})$$

$$\text{So } \frac{kx}{\mu} = 1 - e^{-kt}$$

Make t the subject:

$$e^{-kt} = 1 - \frac{kx}{\mu}$$

$$t = -\frac{1}{k} \ln\left(1 - \frac{kx}{\mu}\right)$$

Substitute this into y to get

$$y = \frac{kw+g}{k^2} \left(1 - e^{\ln\left(1 - \frac{kx}{\mu}\right)}\right) - \frac{g}{k} \left(-\frac{1}{k} \ln\left(1 - \frac{kx}{\mu}\right)\right)$$

$$y = \frac{kw+g}{k^2} \left(1 - \left(1 - \frac{kx}{\mu}\right)\right) + \frac{g}{k^2} \ln\left(1 - \frac{kx}{\mu}\right)$$

$$y = \frac{kw+g}{k^2} \left(\frac{kx}{\mu}\right) + \frac{g}{k^2} \ln\left(1 - \frac{kx}{\mu}\right)$$

$$y = \left(\frac{kw+g}{k\mu}\right)x + \frac{g}{k^2} \ln\left(1 - \frac{kx}{\mu}\right), \text{ as required.}$$

(iv) Substitution of the numerical values gives

$$y = \left(\frac{0.2 \times 6 + 10}{0.2 \times 9}\right) \times 9 + \frac{10}{(0.2)^2} \ln\left(1 - \frac{0.2 \times 9}{9}\right)$$

$$y = 56 + (-55.7859)$$

$y \approx 0.214 \text{ m}$ and $0.214 > 0.15$ so
the particle will pass over the wall.

Q14 (b)

(i) Proof by contrapositive:

Suppose k is odd.

Let $k = 2a + 1$, where a is an integer.

$$\text{Then } k^4 = (2a + 1)^4$$

$$= (4a^2 + 4a + 1)(4a^2 + 4a + 1)$$

$$= 16a^4 + 16a^3 + 4a^2 + 16a^3 + 16a^2 + 4a + 4a^2 + 4a + 1$$

$$= 16a^4 + 32a^3 + 8a^2 + 16a^2 + 8a + 1$$

$$= 2[8a^4 + 16a^3 + 12a^2 + 4a] + 1$$

$$= 2j + 1, \text{ which is odd.}$$

So if k is odd, then k^4 will be odd.

Therefore if k^4 is even, then k is even.

(ii) Proof by contradiction:

Suppose that $\sqrt[4]{8}$ is rational.

Let $\sqrt[4]{8} = \frac{p}{q}$ where p and q are integers with no common factors.

$$\text{Then } 8 = \frac{p^4}{q^4}$$

$$p^4 = 8q^4$$

So p^4 is even

$\therefore p$ is even (from (i))

Let $p = 2m$, m an integer

$$(2m)^4 = 8q^4$$

$$16m^4 = 8q^4$$

$$\therefore q^4 = 2m^4$$

$$\therefore q^4 \text{ is even}$$

$$\therefore q \text{ is even (from (i))}$$

Contradiction!

Both p and q cannot be even as they have no common factors.
Hence $\sqrt[4]{8}$ is irrational.

(c)

Consider the sign of LHS - RHS.

$$\text{LHS} - \text{RHS} = \frac{1}{a} + \frac{1}{b} - \frac{1}{2}$$

$$= \frac{a+b}{ab} - \frac{1}{2}$$

$$= \frac{2(a+b) - ab}{2ab}$$

but $b = 8 - a$, so

$$\text{LHS} - \text{RHS} = \frac{2(a+8-a) - a(8-a)}{2a(8-a)}$$

$$= \frac{16 - 8a + a^2}{2a(8-a)}$$

$$= \frac{(a-4)^2}{2a(8-a)}$$

The numerator is always ≥ 0 .

The denominator is $2ab > 0$ because $a, b \in \mathbb{R}^+$.

Hence $\text{LHS} - \text{RHS} \geq 0$

Hence $\frac{1}{a} + \frac{1}{b} \geq \frac{1}{2}$ as required.

or

$$\text{let } \frac{1}{a} + \frac{1}{b} < \frac{1}{2}$$

$$\therefore \frac{b+a}{ab} < \frac{1}{2}$$

$$\therefore \frac{8}{ab} < \frac{1}{2}$$

$$\therefore 16 < ab \quad \text{since } a, b \text{ are positive}$$

$$\therefore a(8-a) > 16$$

$$\therefore 8a - a^2 > 16$$

$$\therefore a^2 - 8a + 16 < 0$$

$$\therefore (a-4)^2 < 0 \quad \text{contradiction!}$$

$$\therefore \frac{1}{a} + \frac{1}{b} \geq \frac{1}{2}$$

$$\text{Q15, (a)} \int \frac{dx}{x\sqrt{1-x}}$$

$$\text{let } 1-x = u^2$$

$$-dx = 2u du$$

$$dx = -2u du$$

$$I = \int \frac{-2u du}{(1-u^2)\sqrt{u^2}}$$

$$= -2 \int \frac{du}{1-u^2}$$

$$\text{let } \frac{1}{1-u^2} = \frac{A}{1-u} + \frac{B}{1+u}$$

$$1 = A(1+u) + B(1-u)$$

$$A = \frac{1}{2} \text{ and } B = \frac{1}{2}$$

$$I = -2 \int \frac{\frac{1}{2}}{1-u} + \frac{\frac{1}{2}}{1+u} du$$

$$I = - \int \frac{1}{1-u} + \frac{1}{1+u} du$$

$$I = \int \frac{-1}{1-u} - \frac{1}{1+u} du$$

$$I = \ln|1-u| - \ln|1+u|$$

$$I = \ln \left| \frac{1-u}{1+u} \right|$$

$$I = \ln \left| \frac{1-\sqrt{1-x}}{1+\sqrt{1-x}} \right| + C$$

or

$$\text{let } x = \sin^2 \theta$$

$$\therefore dx = 2 \sin \theta \cos \theta d\theta$$

$$\therefore I = \int \frac{2 \sin \theta \cos \theta d\theta}{\sin^2 \theta \sqrt{1-\sin^2 \theta}}$$

$$= 2 \int \frac{\sin \theta \cos \theta d\theta}{\sin^2 \theta \cos \theta}$$

$$= 2 \int \sec \theta d\theta$$

$$= 2 \int \frac{\sec \theta (\sec \theta - \cot \theta) d\theta}{\sec \theta - \cot \theta}$$

$$= 2 \ln |\sec \theta - \cot \theta| + C$$

$$= 2 \ln \left| \frac{1}{\sin \theta} - \frac{\sqrt{1-x}}{\sin \theta} \right| + C$$

$$= 2 \ln \left| \frac{1-\sqrt{1-x}}{\sin \theta} \right| + C$$



(b) $z = e^{i\theta}$

(i) $z = \cos \theta + i \sin \theta$

$z^n = (\cos \theta + i \sin \theta)^n$ — (1)

$= \cos(n\theta) + i \sin(n\theta)$ by de Moivre

$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$

$= \cos(n\theta) - i \sin(n\theta)$ — (2)

(1) + (2) gives

$z^n + z^{-n} = \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta)$
 $= 2 \cos(n\theta)$, as required.

(ii) By the binomial theorem,

$(z + \frac{1}{z})^5 = z^5 + 5z^4(\frac{1}{z}) + 10z^3(\frac{1}{z})^2 + 10z^2(\frac{1}{z})^3 + 5z(\frac{1}{z})^4 + (\frac{1}{z})^5$
 $= z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5}$
 $= (z^5 + \frac{1}{z^5}) + 5(z^3 + \frac{1}{z^3}) + 10(z + \frac{1}{z})$, as required.

(iii) To find $\int \cos^5 \theta d\theta$.

Now $z + \frac{1}{z} = 2 \cos \theta$

$(z + \frac{1}{z})^5 = (2 \cos \theta)^5 = 32 \cos^5 \theta$

$\therefore 32 \cos^5 \theta = 2 \cos 5\theta + 5(2 \cos 3\theta) + 10(2 \cos \theta)$

$\cos^5 \theta = \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta$

$\therefore \int \cos^5 \theta d\theta = \int \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta d\theta$

$= \frac{1}{80} \sin 5\theta + \frac{5}{48} \sin 3\theta + \frac{5}{8} \sin \theta + C$

(c) When $n=2$,

LHS = $\frac{d^2}{dx^2} (x^{\frac{1}{2}})$

$= \frac{d}{dx} (\frac{1}{2} x^{-\frac{1}{2}})$

$= -\frac{1}{4} x^{-\frac{3}{2}}$

RHS = $\frac{(-1) \times 2!}{2^3 \times 1!} x^{-\frac{3}{2}}$

$= -\frac{2}{8} x^{-\frac{3}{2}}$

$= -\frac{1}{4} x^{-\frac{3}{2}}$

LHS = RHS when $n=2$.

The statement is true when $n=2$.

Next, assume that the statement is true when $n=k$, i.e., assume

that $\frac{d^k}{dx^k} (x^{\frac{1}{2}}) = \frac{(-1)^{k-1} (2k-2)!}{2^{2k-1} (k-1)!} x^{-\frac{(2k-1)}{2}}$

and then try to prove that

$\frac{d^{k+1}}{dx^{k+1}} (x^{\frac{1}{2}}) = \frac{(-1)^k (2k)!}{2^{2k+1} (k)!} x^{-\frac{(2k+1)}{2}}$

Now LHS = $\frac{d^{k+1}}{dx^{k+1}} (x^{\frac{1}{2}})$

$= \frac{d}{dx} \left(\frac{d^k}{dx^k} (x^{\frac{1}{2}}) \right)$

$= \frac{d}{dx} \left(\frac{(-1)^{k-1} (2k-2)!}{2^{2k-1} (k-1)!} x^{-\frac{(2k-1)}{2}} \right)$

$= \frac{(-1)^{k-1} (2k-2)!}{2^{2k-1} (k-1)!} \times \frac{-(2k-1)}{2} \times x^{-\frac{(2k-1)}{2} - 1}$

$$\text{LHS} = \frac{(-1)^{k-1} (2k-2)! (-1)(2k-1)}{2^{2k-1} (k-1)! \times 2} \times \frac{-2k-1}{2}$$

$$= \frac{(-1)^k (2k-1)!}{2^{2k} (k-1)!} \times \frac{-(2k+1)}{2}$$

Now multiply the numerator and denominator of the coefficient by $2k$

$$= \frac{(-1)^k (2k-1)! 2k}{2^{2k} (k-1)! 2k} \times \frac{-(2k+1)}{2}$$

$$= \frac{(-1)^k (2k)!}{2^{2k+1} (k)!} \times \frac{-(2k+1)}{2}$$

$$= \text{RHS}$$

Hence the statement is true by mathematical induction.

(d) Proof by contradiction:

Suppose that there is a positive prime number p such that

$$25m^2 + 30m + 9 = n^2 + p$$

$$\text{Then } p = 25m^2 + 30m + 9 - n^2$$

$$p = (5m+3)^2 - n^2$$

$$p = (5m+3-n)(5m+3+n)$$

Notice that $5m+3-n < 5m+3+n$.

Since p is prime then

$$5m+3-n = 1$$

$$\text{and } 5m+3+n = p$$

$$\text{Then } p+1 = 10m+6$$

$$p = 10m+5$$

$$p = 5(2m+1)$$

$$\text{Now } 2m+1 \geq 3$$

$\therefore p = 5(2m+1)$ is a contradiction!

because p cannot be prime and a product of

5 and an integer greater than or equal to 3.

Hence, there is no prime number p such that

$$25m^2 + 30m + 9 = n^2 + p.$$

Q16, (a) (i)

$$\begin{aligned} \text{Now } \vec{CB} &= \vec{b} - \vec{c} \\ \vec{CE} &= \frac{3}{4}(\vec{CB}) \\ &= \frac{3}{4}(\vec{b} - \vec{c}) \\ \vec{CD} &= \frac{1}{4}(-\vec{c}) \\ &= -\frac{1}{4}\vec{c} \end{aligned} \quad \left. \begin{aligned} \vec{DE} &= \vec{CE} - \vec{CD} \\ &= \frac{3}{4}(\vec{b} - \vec{c}) - (-\frac{1}{4}\vec{c}) \\ &= \frac{3}{4}\vec{b} - \frac{3}{4}\vec{c} + \frac{1}{4}\vec{c} \\ &= \frac{3}{4}\vec{b} - \frac{1}{2}\vec{c}, \text{ as required.} \end{aligned} \right\}$$

(ii) To find the ratio of AF:BF

$$\begin{aligned} \text{Let } \vec{AF} &= \lambda \vec{AB} = \lambda \vec{b} \\ \text{Let } \vec{DF} &= \mu \vec{DE} = \mu \left(\frac{3}{4}\vec{b} - \frac{1}{2}\vec{c} \right). \end{aligned}$$

$$\begin{aligned} \text{Now } \vec{AF} &= \vec{AD} + \vec{DF} \\ &= \frac{3}{4}\vec{c} + \mu \left(\frac{3}{4}\vec{b} - \frac{1}{2}\vec{c} \right) \\ &= \frac{3}{4}\vec{c} + \frac{3}{4}\mu\vec{b} - \frac{\mu}{2}\vec{c} \\ &= \frac{3-2\mu}{4}\vec{c} + \frac{3\mu}{4}\vec{b} \end{aligned}$$

$$\text{but } \vec{AF} = \lambda \vec{b} \text{ so}$$

$$\begin{aligned} \frac{3-2\mu}{4} &= 0 & \text{and } \frac{3\mu}{4} &= \lambda \\ \therefore \mu &= \frac{3}{2} & \therefore \lambda &= \frac{9}{8} \end{aligned}$$

Sub $\lambda = \frac{9}{8}$ into

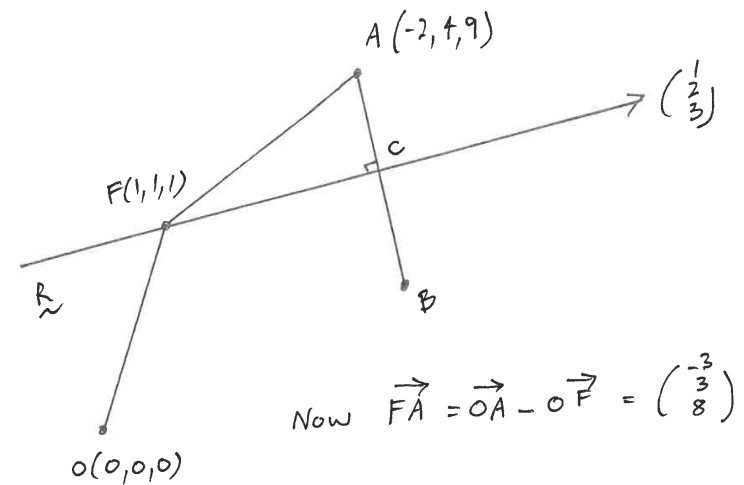
$$\vec{AF} = \lambda \vec{AB}$$

$$8\vec{AF} = 9\vec{AB}$$

$$8\vec{AF} = 9(\vec{AF} - \vec{BF})$$

$$\therefore 9\vec{BF} = \vec{AF} \Rightarrow \therefore AF:BF = 9:1$$

(b)



$$\text{Now } \vec{FA} = \vec{OA} - \vec{OF} = \begin{pmatrix} -3 \\ 3 \\ 8 \end{pmatrix}$$

$\vec{FC}$ is the projection of $\vec{FA}$ onto r .

$$\begin{aligned} \text{Hence } \vec{FC} &= \frac{\begin{pmatrix} -3 \\ 3 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}}{(\sqrt{14})^2} r \\ &= \frac{-3+6+24}{14} \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} \\ &= \frac{27}{14} \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
 \text{Now } \vec{AC} &= \vec{FC} - \vec{FA} \\
 &= \frac{27}{14} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} -3 \\ 3 \\ 8 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{69}{14} \\ \frac{6}{7} \\ -\frac{31}{14} \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{And } \vec{OB} &= \vec{OA} + 2 \times \vec{AC} \\
 &= \begin{pmatrix} -2 \\ 4 \\ 9 \end{pmatrix} + 2 \begin{pmatrix} \frac{69}{14} \\ \frac{6}{7} \\ -\frac{31}{14} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{55}{7} \\ \frac{40}{7} \\ \frac{32}{7} \end{pmatrix}
 \end{aligned}$$

$$\therefore B = \left(\frac{55}{7}, \frac{40}{7}, \frac{32}{7} \right)$$

(c)

$$\begin{aligned}
 \text{(i)} \quad &\int \sin^4 x \cos^3 x \, dx \\
 &= \int \sin^4 x (1 - \sin^2 x) \cos x \, dx \\
 &= \int \sin^4 x \cos x - \sin^6 x \cos x \, dx \\
 &= \frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad &\int \sin^m x \cos^n x \, dx = I \\
 &I = \int (\sin^{m-1} x \cos x) (\cos^{n-1} x \sin x) \, dx
 \end{aligned}$$

Note: breaking up $\cos^n x = \cos x \cos^{n-1} x$ is necessary. Breaking up $\sin^m x$ was unnecessary though! might save a few lines not doing it.

$$\begin{aligned}
 &= \left(\frac{1}{m} \sin^m x \right) (\cos^{n-1} x \sin x) - \int \frac{1}{m} \sin^m x \times \\
 &\quad \left[\cos^{n-1} x \cdot \cos x + \sin x (n-1) \cos^{n-2} x (-\sin x) \right]
 \end{aligned}$$

$$= \frac{1}{m} \sin^{m+1} x \cos^{n-1} x - \int \frac{1}{m} \sin^m x \left[\cos^n x - (n-1) \cos^{n-2} x \sin^2 x \right] dx$$

$$= \frac{1}{m} \sin^{m+1} x \cos^{n-1} x - \int \frac{1}{m} \sin^m x \cos^n x - \frac{1}{m} (n-1) \sin^{m+2} x \cos^{n-2} x \, dx$$

$$= \frac{1}{m} \sin^{m+1} x \cos^{n-1} x - \frac{1}{m} I + \frac{n-1}{m} \int \sin^{m+2} x \cos^{n-2} x \, dx$$

$$\therefore I + \frac{1}{m} I = \frac{1}{m} \sin^{m+1} x \cos^{n-1} x + \frac{n-1}{m} \int \sin^{m+2} x \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$\therefore \left(\frac{m+1}{m} \right) I = \frac{1}{m} \sin^{m+1} x \cos^{n-1} x + \frac{n-1}{m} \int \sin^{m+2} x \cos^{n-2} x - \sin^m x \cos^n x \, dx$$

$$\therefore \left(\frac{m+1}{m} \right) I = \frac{1}{m} \sin^{m+1} x \cos^{n-1} x + \frac{n-1}{m} \int \sin^{m+2} x \cos^{n-2} x \, dx - \frac{n-1}{m} I$$

$$\therefore \left(\frac{m+1}{m} \right) I + \left(\frac{n-1}{m} \right) I = \frac{1}{m} \sin^{m+1} x \cos^{n-1} x + \frac{n-1}{m} \int \sin^{m+2} x \cos^{n-2} x \, dx$$

$$\left(\frac{m+n}{m}\right) I = \frac{1}{m} \sin^{m+1} x \cos^{n-1} x + \frac{n-1}{m} \int \sin^m x \cos^{n-2} x dx$$

$$\therefore I = \frac{1}{m+n} \sin^{m+1} x \cos^{n-1} x + \frac{n-1}{m+n} \int \sin^m x \cos^{n-2} x dx$$

$$\therefore I = \frac{\sin^{m+1} x \cos^{n-1} x}{m+n} + \frac{n-1}{m+n} \int \sin^m x \cos^{n-2} x dx$$

as required.

$$(iii) \int \sin^4 x \cos^3 x dx$$

$$= \frac{\sin^5 x \cos^2 x}{7} + \frac{2}{7} \int \sin^4 x \cos x dx$$

$$= \frac{\sin^5 x \cos^2 x}{7} + \frac{2}{7} \cdot \frac{1}{5} \sin^5 x + c$$

$$= \frac{\sin^5 x (1 - \sin^2 x)}{7} + \frac{2}{35} \sin^5 x + c$$

$$= \frac{\sin^5 x}{7} - \frac{\sin^7 x}{7} + \frac{2}{35} \sin^5 x + c$$

$$= \frac{7}{35} \sin^5 x - \frac{1}{7} \sin^7 x + c$$

$$= \frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x + c$$

which confirms the answer from part (i).

END OF SOLUTIONS.